

A radial-envelope extension for extremal general affine surface area

Candidate partial result for arXiv:2103.00294

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Status. Candidate partial result, likely valid, pending human review.

1 Source problem and claimed scope

Hoehner studies extremal general affine surface areas for centered convex bodies and function classes determined by global monotonicity of $\varphi(t)/\sqrt{t}$ [1]. At the end of Section 7.4 (page 21), the paper asks to “obtain the results of this article for a richer class of functions than $\text{Conc}^-(0, \infty) \cup \text{Conc}^+(0, \infty)$.”

function $G_n(t) = \psi(t^{n+1})$ for $t \in (0, \infty)$ is convex. Then

$$\text{as}_\psi(K) \geq \text{as}_\psi(\text{vrad}(K)B_n).$$

Moreover, if $G_n(\cdot)$ is strictly convex, then equality holds if and only if K is an origin-symmetric ellipsoid.

Thus, one could remove the assumption that the bodies in $\mathcal{C}_{\text{in}}(K)$ or $\mathcal{C}_{\text{out}}(K)$ have centroid at the origin by using Lemmas 7.1 and 7.2 in all of the proofs instead of Lemmas 2.1 and 2.2, but the downside is then one captures fewer functions from $\text{Conc}(0, \infty)$ and $\text{Conv}(0, \infty)$. Furthermore, the assumption in the definitions of $\mathcal{C}_{\text{in}}(K)$ and $\mathcal{C}_{\text{out}}(K)$ that the bodies have centroid at the origin is tacitly assumed in the definition of the extremal L_p affine surface areas in [18]. This is because the definition (39) of L_p affine surface area is stated for convex bodies with centroid at the origin. Ultimately, it is an open problem to obtain the results of this article for a richer class of functions than $\text{Conc}^-(0, \infty) \cup \text{Conc}^+(0, \infty)$.

This note gives a partial answer. It strictly enlarges the function class for which the inner maximal functional is finite, attained, and continuous, and it replaces the source’s endpoint radial estimate by a sharp radial envelope. It does not extend every theorem of the source paper.

2 Definitions

Let B_n be the Euclidean unit ball in $\mathbb{R}^n$. For a convex body K , write

$$\text{vrad}(K) = \left(\frac{|K|}{|B_n|} \right)^{1/n}.$$

All convex bodies below have centroid at the origin. Let $\mathcal{C}_{\text{in}}(K)$ denote the convex bodies $C \subset K$ whose centroid is the origin.

Let $\text{Conc}(0, \infty)$ be the source class of positive concave functions $\varphi : (0, \infty) \rightarrow (0, \infty)$ satisfying $\varphi(0) = 0$ and $\varphi(t)/t \rightarrow 0$ as $t \rightarrow \infty$. Its general affine surface area is denoted by as_φ . Define

$$\text{IS}_\varphi(K) = \sup_{C \in \mathcal{C}_{\text{in}}(K)} \text{as}_\varphi(C).$$

The two source facts used below are the affine isoperimetric inequality

$$\text{as}_\varphi(C) \leq \text{as}_\varphi(\text{vrad}(C)B_n) \quad (1)$$

for centered C , and upper semicontinuity of as_φ under Hausdorff convergence [2, 1]. On a Euclidean ball,

$$\text{as}_\varphi(rB_n) = r^n \varphi(r^{-2n}) |\partial B_n| = n|B_n| \frac{\varphi(t)}{\sqrt{t}}, \quad t = r^{-2n}. \quad (2)$$

Put

$$h_\varphi(t) = \frac{\varphi(t)}{\sqrt{t}}, \quad L_\varphi = \limsup_{t \rightarrow \infty} h_\varphi(t), \quad H_\varphi(a) = \sup_{t \geq a} h_\varphi(t).$$

Call φ *tail-gapped* when

$$\begin{aligned} L_\varphi &< \infty, \\ H_\varphi(a) &> L_\varphi \quad \text{for every } a > 0. \end{aligned} \quad (3)$$

3 Proof intuition

The source proof controls an inscribed body C only through its volume radius. Formula (2) shows that the correct one-dimensional object is not an endpoint value of h_φ , but its tail envelope H_φ . Finite tail limsup makes this envelope finite. The strict tail gap then prevents a maximizing sequence from collapsing: a degenerate sequence can have limiting affine surface area at most $n|B_n|L_\varphi$, while a sufficiently small centered ball inside K has strictly larger value. Once collapse is excluded, Blaschke selection and upper semicontinuity give an extremizer. Continuity follows separately from inclusion monotonicity and the elementary dilation estimate, so it does not require global monotonicity of h_φ .

4 Radial-envelope theorem

Theorem 1. *Let $\varphi \in \text{Conc}(0, \infty)$ be tail-gapped. Then for every centered convex body $K \subset \mathbb{R}^n$:*

1. $0 < \text{IS}_\varphi(K) < \infty$, and some $C_{\max} \in \mathcal{C}_{\text{in}}(K)$ satisfies $\text{IS}_\varphi(K) = \text{as}_\varphi(C_{\max})$;
2. the sharp radial-envelope bound holds:

$$\text{IS}_\varphi(K) \leq n|B_n| H_\varphi(\text{vrad}(K)^{-2n}) \quad (4)$$

3. equality in (4) holds when $K = RB_n$ for every $R > 0$;
4. $K \mapsto \text{IS}_\varphi(K)$ is continuous on centered convex bodies in the Hausdorff metric.

Moreover, every function in the source class $\text{Conc}^-(0, \infty)$ is tail-gapped.

Proof. Set $R = \text{vrad}(K)$ and $a = R^{-2n}$. For $C \in \mathcal{C}_{\text{in}}(K)$, volume inclusion gives $\text{vrad}(C) \leq R$. With $t_C = \text{vrad}(C)^{-2n} \geq a$, equations (1) and (2) give

$$\text{as}_\varphi(C) \leq n|B_n| h_\varphi(t_C) \leq n|B_n| H_\varphi(a).$$

Because h_φ is continuous and has finite tail limsup, $H_\varphi(a) < \infty$. This proves finiteness and (4). Positivity follows because K contains a centered ball of positive radius and φ is positive.

Let $S = \text{IS}_\varphi(K)$ and choose $C_j \in \mathcal{C}_{\text{in}}(K)$ with $\text{as}_\varphi(C_j) \rightarrow S$. By Blaschke's selection theorem [3], a subsequence converges in the Hausdorff metric to a compact convex set $C_0 \subset K$. Suppose that C_0 has empty interior. Then $|C_j| \rightarrow 0$, hence $t_j = \text{vrad}(C_j)^{-2n} \rightarrow \infty$. The preceding bound gives

$$\limsup_j \text{as}_\varphi(C_j) \leq n|B_n| \limsup_j h_\varphi(t_j) \leq n|B_n|L_\varphi. \quad (5)$$

Choose $\rho > 0$ with $\rho B_n \subset K$. By $H_\varphi(\rho^{-2n}) > L_\varphi$, there is a $t \geq \rho^{-2n}$ with $h_\varphi(t) > L_\varphi$. The ball rB_n , $r = t^{-1/(2n)} \leq \rho$, belongs to $\mathcal{C}_{\text{in}}(K)$, and therefore

$$S \geq \text{as}_\varphi(rB_n) = n|B_n|h_\varphi(t) > n|B_n|L_\varphi,$$

contradicting (5). Thus C_0 has nonempty interior. Centroid continuity gives $g(C_0) = 0$, so $C_0 \in \mathcal{C}_{\text{in}}(K)$. Upper semicontinuity now yields

$$S = \limsup_j \text{as}_\varphi(C_j) \leq \text{as}_\varphi(C_0) \leq S.$$

Hence the supremum is attained.

For $K = RB_n$, every $t \geq R^{-2n}$ is realized by the centered ball $t^{-1/(2n)}B_n \subset RB_n$. Formula (2) supplies the reverse inequality to (4), proving equality. Notice also that the supremum defining $H_\varphi(a)$ is attained at a finite t : otherwise a maximizing sequence escaping to infinity would force $H_\varphi(a) \leq L_\varphi$, contrary to (3).

It remains to prove continuity. The functional IS_φ is increasing under inclusion. Also, for $c \geq 1$, the source dilation estimate $\text{as}_\varphi(cC) \leq c^n \text{as}_\varphi(C)$ gives

$$\text{IS}_\varphi(cK) \leq c^n \text{IS}_\varphi(K). \quad (6)$$

Suppose centered K_j converges to centered K in Hausdorff distance. Since the origin is interior to K , standard support-function inclusions give numbers $\delta_j, \eta_j \downarrow 0$ such that, for all large j ,

$$K_j \subset (1 + \delta_j)K, \quad K \subset (1 + \eta_j)K_j.$$

Monotonicity and (6) imply

$$(1 + \eta_j)^{-n} \text{IS}_\varphi(K) \leq \text{IS}_\varphi(K_j) \leq (1 + \delta_j)^n \text{IS}_\varphi(K),$$

and continuity follows.

Finally, if $\varphi \in \text{Conc}^-(0, \infty)$, then h_φ is positive and strictly decreasing. Its finite tail limit is L_φ , and $H_\varphi(a) = h_\varphi(a) > L_\varphi$ for every $a > 0$. Thus the entire source decreasing-ratio class is contained in the tail-gapped class. $\square$

5 A strict extension: the logarithmic function

Corollary 2. *Let $\varphi(t) = \log(1 + t)$. Then φ is tail-gapped but belongs to neither $\text{Conc}^-(0, \infty)$ nor $\text{Conc}^+(0, \infty)$. There is a unique $t_* > 1$ satisfying*

$$\log(1 + t_*) = \frac{2t_*}{1 + t_*}, \quad (7)$$

and $t_* = 3.9215536346\dots$. If $h(t) = \log(1 + t)/\sqrt{t}$, then

$$H(a) = \begin{cases} h(t_*), & 0 < a \leq t_*, \\ h(a), & a \geq t_*. \end{cases}$$

Consequently Theorem 1 applies, and in particular

$$\text{IS}_\varphi(B_n) = n|B_n|h(t_*) > n|B_n|h(1) = \text{as}_\varphi(B_n).$$

Proof. The function φ is positive and concave, with $\varphi(0) = 0$ and $\varphi(t)/t \rightarrow 0$. Direct differentiation shows that the sign of $h'(t)$ is the sign of

$$q(t) = \frac{2t}{1+t} - \log(1+t).$$

Here $q(0) = 0$ and

$$q'(t) = \frac{1-t}{(1+t)^2}.$$

Thus q increases on $(0, 1)$, decreases on $(1, \infty)$, and tends to $-\infty$. Since $q(1) = 1 - \log 2 > 0$, it has a unique positive zero $t_* > 1$. Therefore h increases up to t_* and decreases afterwards. In particular it is globally nonmonotone. Since $h(t) \rightarrow 0$ as $t \rightarrow \infty$, it is tail-gapped and the displayed envelope follows. Equation (7) is $q(t_*) = 0$; the reported decimal is checked reproducibly by `code/check_log_example.py` and is not used in the proof. $\square$

Remark 3 (Repair of the source non-collapse step). *The proof of the source existence lemma writes that strict decrease of h_φ makes the right side of (5) tend to zero. This is stronger than the definition. For $\varphi(t) = \sqrt{t} + t^{1/4}$, one has $h_\varphi(t) = 1 + t^{-1/4} \downarrow 1$. The source conclusion is nevertheless recovered by the strict tail gap: every finite $h_\varphi(t)$ lies strictly above its tail limit. Theorem 1 isolates exactly the compactness mechanism needed here and also admits nonmonotone ratios.*

6 Verification and limitations

The proof was checked step by step against the following failure modes. The affine isoperimetric bound is applied only to centered bodies; the tail gap is imposed for every $a > 0$, so it works for bodies of arbitrary inradius; centroid continuity is invoked only after degeneration is excluded; and the continuity argument uses only finite values of IS_φ . If $L_\varphi = \infty$, arbitrarily small centered balls can make $\text{IS}_\varphi(K) = \infty$, so tail boundedness is a real restriction. Without the strict gap, this compactness proof need not prevent degeneration; no necessity assertion is made.

The original problem is broad. The theorem above extends existence, attainment, continuity, and the radial affine-isoperimetric estimate for the inner maximal functional. The logarithmic example shows that the source’s ball identity and equality theory cannot extend unchanged. A full extension of the Blaschke-Santaló theorem remains open.

Bounded novelty check. The four local run indexes were searched by arXiv id, title, and core terms. Exact-title and exact-phrase arXiv/web searches were run through 19 July 2026. The journal page lists three citing works (a 2023 volume-product survey and two 2025 convex-geometry papers); none states the tail-envelope theorem. The 2024 preprint *Extremal affine surface areas in a functional setting* (arXiv:2402.16130) treats a different functional construction and fixed power parameters. This bounded search found no prior statement of the present extension, but absolute novelty is not certified.

Human review recommendation. Send to a convex-geometric analyst. The highest-value checks are (5), the exact hypotheses of the affine isoperimetric and upper-semicontinuity inputs, and whether this theorem should be described as a partial answer to the source’s deliberately broad phrase “the results.”

References

- [1] S. Hoehner, *Extremal general affine surface areas*, J. Math. Anal. Appl. 505 (2022), 125506; arXiv:2103.00294.
- [2] M. Ludwig, *General affine surface areas*, Adv. Math. 224 (2010), 2346–2360.
- [3] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014.